

Seyedmahdiar Ziaolhagh

Aerospace Engineer | Experimental Systems | Mechanical Design | Controls

Lethbridge, Alberta, Canada	ziaolhagh@uleth.ca 403-929-1520	LinkedIn
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PROFILE

Aerospace engineer and graduate researcher combining mechanical design, controls, experimental systems, vacuum and cryogenic testing, optical metrology, vibration analysis, scientific programming, and engineering validation. Experienced in translating physical requirements into CAD, simulation, instrumentation, test procedures, and evidence-based design decisions. Authorized to work in Canada under an open work permit.

TECHNICAL SKILLS

CAD AND MANUFACTURING	CATIA V5, Siemens NX, SolidWorks, Knowledgeware, Generative Shape Design, sheet metal, assemblies, GD&T, engineering drawings, 3D printing, laser cutting
SIMULATION AND CONTROLS	MATLAB/Simulink, Abaqus, ANSYS STK, PID and state-space control, structural and thermal analysis, FEA, vibration analysis, particle swarm optimization
PROGRAMMING AND DATA	Python, NumPy, SciPy, Pandas, Matplotlib, h5py, signal processing, FFT, uncertainty analysis, Monte Carlo methods, automated data processing
LABORATORY AND METROLOGY	PhaseCam 4030, 4Sight Focus, interferometry, vacuum systems, residual gas analyzer, cryogenic testing, composite testing, mechanical testing
ELECTRONICS AND INTEGRATION	Sensors, oscilloscopes, synchronized DAQ, Arduino/AccelStepper, stepper drives, LTspice, RGASoft, UAV integration, experimental setup design

EDUCATION

SEPT. 2024 - PRESENT	Master of Science in Experimental Physics University of Lethbridge Lethbridge, Alberta, Canada Thesis: Evaluation of composite-material mechanical properties in space and stratospheric environments. Research includes cryogenic systems, vacuum instrumentation, vibration testing, and optical metrology.
SEPT. 2019 - FEB. 2024	Bachelor of Engineering in Aerospace Engineering K. N. Toosi University of Technology Tehran, Iran GPA 16.5/20; top 10% of cohort. Thesis: Mineral extraction scenarios from an asteroid via optimization considering rotational dynamics.

PROFESSIONAL EXPERIENCE

JAN. 2025 - PRESENT LETHBRIDGE, CANADA	Research Assistant University of Lethbridge - Designing and validating a custom three-stage vacuum and cryogenic system for mechanical characterization of CFRP at room and cryogenic temperatures. - Developing experimental methods for composite testing in simulated space and stratospheric environments and characterizing outgassing with a residual gas analyzer. - Processed data from more than 300 vibration experiments to estimate mechanical properties and quantify uncertainty. - Developed an automated two-axis interferometric surface profiler with motion control, overlap registration, Python-based stitching, and high-resolution surface reconstruction. - Integrated sensors, motion control, synchronized acquisition, and scientific instrumentation into repeatable laboratory workflows.
JAN. 2025 - MAY 2025 LETHBRIDGE, CANADA	Teaching Assistant - Engineering Physics University of Lethbridge - Delivered laboratory instruction in experimental methods, uncertainty analysis, and scientific reporting. - Mentored students in troubleshooting experiments, interpreting results, and improving technical reports; evaluated laboratory performance and provided technical feedback.
APR. 2023 - NOV. 2024 TEHRAN, IRAN	Mechanical Component Designer Fara-Damay-E-Alborz Ltd. - Designed custom mechanical components and assemblies to customer requirements, functional specifications, and manufacturing constraints. - Produced CAD models and engineering drawings and collaborated with customers to improve manufacturability, performance, and cost.
SEPT. 2023 - DEC. 2023 TEHRAN, IRAN	Teaching Assistant - Dynamics K. N. Toosi University of Technology Led weekly tutorials, created assignments and examinations, graded coursework, and supported students in analytical engineering mechanics.

SELECTED ENGINEERING PROJECTS

CAD-Informed Aileron Actuator Modelling and Control Integrated CATIA hinge-axis inertia ($2.183 \times 10^{-4} \text{ kg m}^2$) into a constrained MATLAB/Simulink PIDF model. Achieved 1.603% overshoot, 0.948 s settling time, and assessed response to a 1.5 N m disturbance.	Aerospace Actuator Bracket Developed a Siemens NX part and constrained assembly, completed preliminary 5 kN static FEA and mesh refinement, and prepared a two-sheet GD&T manufacturing package.
CATIA V5 Parametric Wing Built a Knowledgeware-driven wing with multi-section surfaces, guide curves, structural organization, wing-box definition, and a manufacturing-oriented sheet-metal rib.	Three-Stage Cryogenic Composite Test System Designed a vacuum and cryogenic architecture for CFRP testing at nominal 300 K, 65 K, and 4.2 K stages; integrated mechanical, thermal, vacuum, and instrumentation requirements.

SELECTED ENGINEERING PROJECTS - CONTINUED

Interferometric Surface Profiling System Designed an automated two-axis scan stage and Python stitching workflow for eight overlapping PhaseCam tiles over approximately 21.5 x 11.8 mm. Latest validation achieved 8.28 nm median overlap RMS and 0.922 independent row-profile correlation.	Vibration-Based Young's Modulus Characterization Validated loaded and unloaded cantilever methods with MATLAB/Simulink and experimental data. Compared reference materials and propagated dimensional uncertainty through Monte Carlo analysis.
Asteroid Mining Optimization Developed a particle swarm optimization framework for material extraction from asteroid Bennu while constraining changes in rotational dynamics; evaluated 30% and 70% extraction scenarios.	Undergraduate Aerospace Engineering Portfolio Completed reaction-wheel attitude control, proportional-navigation guidance, turbofan material selection, electrical-harness thermal design, GRACE mission analysis, and UAV development projects.

PUBLICATION AND RESEARCH PRESENTATIONS

JOURNAL PUBLICATION

Mineral Extraction Scenarios From an Asteroid via Optimization Considering Rotational Dynamics
S. Ziaolhagh and M. Jafari-Nadoushan, IEEE Access, 2025. DOI: 10.1109/ACCESS.2025.3615449

CONFERENCE AND RESEARCH PRESENTATIONS

- Characterization of Composite Materials for Space Environments - Canadian Astronomical Society Annual Meeting (CASCA), Montreal, 2026.
- Technology Development for Space-Based Far-Infrared Spectroscopy and Interferometry - CASCA, Montreal, 2026.
- When Your Equipment Lies to You and What to Do About It - University of Lethbridge Summer Research Showcase, 2025.
- Laser Metrology for Material Characterization Experiments - University of Lethbridge Summer Research Showcase, 2025.

LEADERSHIP, TEACHING, AND OUTREACH

2025 - 2026	STEM Outreach Presenter and Laboratory Tour Guide, University of Lethbridge Presented physics applications, demonstrated experiments, and introduced high-school students to instrumentation laboratories and research activities.
MAY - SEPT. 2025	SolidWorks Instructor, University of Lethbridge Designed and delivered beginner-to-advanced SolidWorks instruction for summer students.
SEPT. 2022 - JAN. 2025	Scientific Researcher and Technical Writer, OWJ Aerospace Student Magazine Researched aerospace topics, interviewed specialists, and wrote technical articles on space exploration and technology.

CERTIFICATES, HONOURS, AND LANGUAGES

CERTIFICATES • NASA Open Science 101 Curriculum, 2025. • NSERC SMART CREATE Entrepreneurship Workshop, University of Alberta, 2025. • Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University, 2023. • Commercial Pilot Ground Course, Homa Aviation Center, 2022.	HONOURS AND LANGUAGES • Top 10% of aerospace engineering students during the B.Eng. program. • Third place, World Robot Olympiad qualifiers in Iran; participant at the international event in Greater Noida, India, 2016. • Persian: Native English: Fluent.
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